Investigating Brain-LLM Alignment with Category-Deprived Text

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The question we address is whether the alignment of (Large) Language Models (LMs) to BOLD response in different brain regions can be modulated by training the LM on "category deprived" corpora (i.e. corpora where we have systematically removed nouns, verbs, content words or function words), and which LM layers work best to establish the correlation. Results show that systematically eliminating categories decreases the LM-Brain correlation compared to eliminating the same amount of words picked randomly, that different categories affect partly orthogonal regions and that models and layers differ significantly.

1 Background: LLM-to-Brain alignment studies

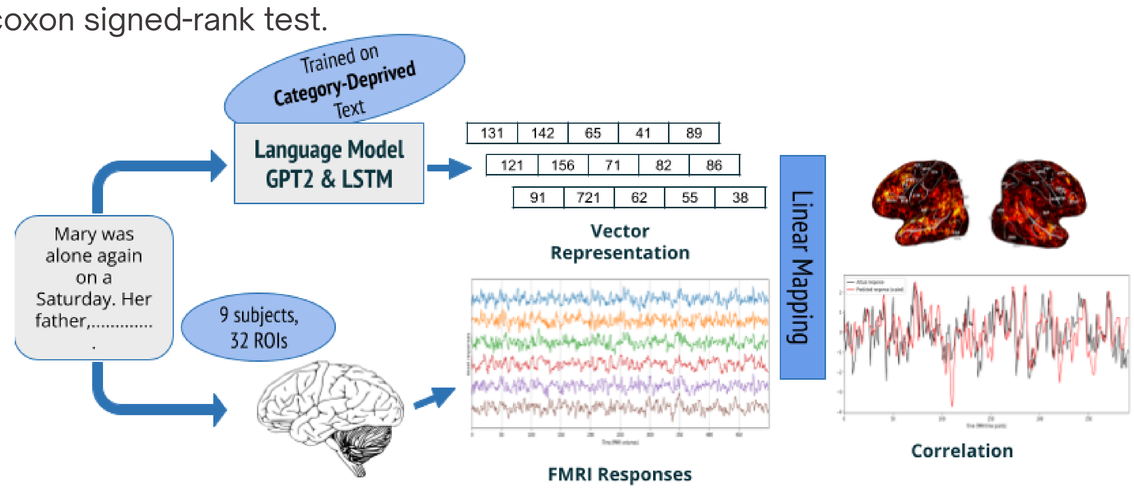
- LMs that have been trained for next word prediction using large text corpora can be leveraged to significantly predict brain activity of subjects that listen to or read text (Jain & Huth, 2018, Toneva & Wehbe, 2019, Schrimpf et al., 2021, Oota et al., 2022).
- However, it remains unclear what linguistic features contribute to this effect (Jawahar et al., 2019, Conneau et al., 2018).
- Many experiments in cognitive neuroscience examine how particular language features are processed by **manipulating the experimental stimulus** and tracking brain region BOLD responses or processing times (Pallier et al., 2011).
- Oota et al. (2022) proposes a *residual approach* to study how eliminating linguistic information from the encodings affects this alignment.
- However, we do not know of attempts of **eliminating linguistic information at the source, from the training an LM receives**. Here we report preliminary results on precisely this approach.
- **We examine how removing different linguistic Part-Of-Speech (PoS) categories affects the correlation between LMs and brain responses.**

2 Experimental design: LM side

- We created multiple versions of a 12M English corpus sampled from Wikipedia, UKWAC and Gutemberg, removing token distinctions within a specific POS category (e.g. all nouns become "NNOUN", or all main verbs, "VVERBM", etc.).
- We then trained from scratch two different character-based LMs (LSTM and GPT2-small) on the original corpus (**LM(O)**), on the various category-corrupted corpora (**LM(O-N**(ouns)**), LM(O-V**(erbs)**), LM(O-C**(content)**), LM(O-F**(unction))
- As control condition, we trained the same models on other versions of **O** where we had *randomly* replaced words with a "WWORD" tag, in the same proportion as the various category-corrupted corpora (**LM(O-R**(andom)**N**(oun), etc.).

3 Experimental design: brain side

- We obtained he texts of spoken narrative stories that were used to collect fMRI responses of subjects while they were listening to these stories (Huth et al., 2016) and fed them to our set of LMs; we collected the representations from the last layer (**LL**), for GPT-2, LSTM, or an intermediate layer (**IL**), for GPT-2.
- we calculate the correlation **C(.)** by learning an encoding model that maps the two representations (see Jain & Huth, 2018). The results were evaluated on held-out test data.
- **we then compared the correlation between LL/IL and the whole brain, and between LL/IL and each of 32 Regions of Interest (ROI)** taken from the SENSAAS atlas (Labache et al., 2019).
- for each ROI, we subtracted the category-condition correlation from the control-condition correlation (e.g. **C(LM(O-RN)) - C(LM(O-N))**), to make sure that drops in correlation were not simply due to the LM receiving a "worse" input.
- The significance of the random/category condition difference was validates via Wilcoxon signed-rank test.



4 Results: whole brain

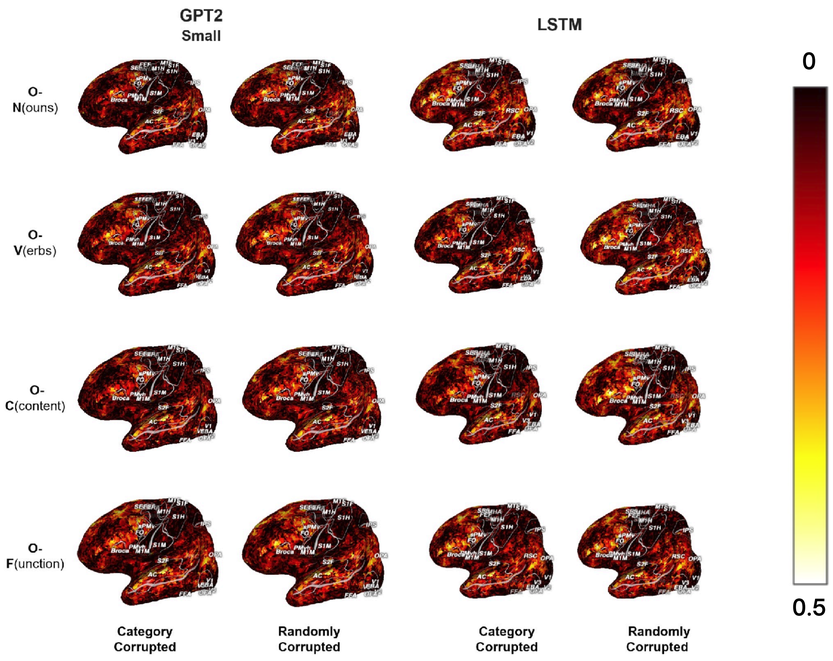


Fig 1: Whole brain alignment correlation maps (original versus category condition) reveal: (i) Stronger effects for Nouns in left anterior temporal and parietal areas canonically associated with lexico-semantic processing (Binder et al. 2009). (ii) Stronger effects for Verbs in frontal and posterior temporal areas that are typically associated with verb processing (Lukic et al., 2021). (iii) Effects for Content and Function Words were somewhat stronger than those associated with the individual ablation of Nouns and Verbs, compatible with greater overall lack of lexico-semantic and syntactic information, and partially reflecting the anatomical distribution of both Noun-less and Verb-less effects.

5 Results: ROI-to-RND Control correlation - ROI-to-Category correlation

Brain ROIs	GPT-2 Layer 12				GPT-2 Layer 6				LSTM			
	No NOUNS	No Verbs	No Cont. Words	No Funct. Words	No NOUNS	No Verbs	No Cont. Words	No Funct. Words	No NOUNS	No Verbs	No Cont. Words	No Funct. Words
Superior Frontal Gyrus	**0.009**	0.005	0.003	0.006	0.003	0.003	0.006	0.006	0.003	0.001	**0.009**	**0.008**
Inferior Frontal Gyrus (Pars Triangularis)	0.004	0.006	0.007	0.005	0.002	-0.004	**0.009**	0.008	0.001	-0.006	0.009	0.008
Inferior Frontal Gyrus (Pars Orbitalis)	0.005	0.006	0.001	0.008	0.003	-0.001	0.003	0.012	0.006	0.004	**0.011**	**0.015**
Inferior Frontal Sulcus	0.008	0.004	0.006	**0.008**	0.004	0.002	**0.011**	0.009	0.005	-0.003	0.006	**0.009**
Anterior Insular Gyrus 2	**0.009**	0.006	**0.010**	**0.011**	0.004	0.004	**0.011**	**0.013**	-0.002	-0.002	0.005	0.003
Anterior Insular Gyrus 3	**0.003**	**0.011**	0.005	**0.011**	0.004	0.000	**0.010**	0.004	0.005	0.000	0.006	**0.012**
Anterior Insular Gyrus 1	**0.009**	0.005	0.004	**0.010**	0.003	0.002	0.006	0.005	0.002	**0.004**	0.009	**0.009**
Precentral Sulcus 3	0.011	**0.016**	-0.002	0.009	-0.002	-0.004	0.007	0.009	0.003	-0.003	0.008	0.013
Precentral Sulcus 4	-0.001	**0.008**	**0.008**	0.006	0.002	0.005	0.007	0.007	0.005	0.001	**0.007**	**0.008**
Supplementary Motor Area (SMA) 1	0.004	0.003	0.002	0.005	0.002	0.006	0.007	0.005	0.001	0.001	**0.013**	**0.010**
Supplementary Motor Area (SMA) 2	0.003	**0.007**	0.001	**0.006**	0.000	0.002	0.005	0.005	0.005	-0.002	0.011	0.008
Paracentral Lobule	0.004	**0.008**	0.007	0.003	0.002	**0.004**	0.008	0.006	-0.002	0.004	**0.008**	**0.009**
Supramarginal Gyrus	0.000	0.003	-0.001	0.000	0.011	-0.002	0.008	0.011	-0.001	-0.003	-0.002	**0.006**
Angular Gyrus	0.009	0.006	0.004	**0.009**	0.003	0.005	0.006	**0.010**	0.000	0.002	0.010	**0.008**
Precuneus Gyrus	**0.010**	**0.008**	0.000	0.007	0.001	0.001	0.011	0.006	0.003	-0.003	**0.013**	0.004
Posterior Cingulate Gyrus	0.013	0.002	0.004	0.002	-0.003	0.012	0.015	0.006	0.002	0.004	**0.019**	0.001
Superior Temporal Sulcus 1	0.011	**0.016**	0.006	0.014	0.003	0.003	**0.012**	0.009	0.004	-0.005	0.010	**0.010**
Superior Temporal Sulcus 3	0.002	0.007	0.004	**0.009**	0.003	0.006	0.003	0.005	0.005	-0.003	-0.001	**0.007**
Superior Temporal Sulcus 2	0.005	0.012	0.008	0.007	0.005	**0.006**	**0.014**	0.006	0.005	0.005	**0.012**	**0.013**
Superior Temporal Sulcus 4	-0.003	0.008	0.005	0.007	0.005	-0.002	0.002	**0.014**	0.005	-0.005	-0.001	**0.008**
Middle Temporal Gyrus 3	-0.006	0.008	-0.002	0.004	0.006	0.006	0.008	**0.007**	-0.003	0.000	-0.002	**0.009**
Inferior Temporal Gyrus	0.003	0.008	0.006	**0.020**	0.002	-0.002	0.005	0.012	0.002	0.002	0.001	**0.010**
Fusiform Gyrus	-0.004	0.008	0.002	0.005	0.003	**0.007**	0.008	0.009	0.008	-0.003	0.006	**0.009**
Hippocampus	**0.006**	0.006	0.005	0.009	-0.001	0.001	0.008	**0.008**	0.001	0.002	**0.013**	**0.009**
Parahippocampal Gyrus	**0.009**	0.003	0.003	0.004	0.004	0.006	0.015	0.006	0.004	-0.003	0.007	0.005
Amygdala Nucleus	**0.012**	0.003	0.003	**0.010**	0.003	0.002	0.006	0.008	0.000	0.000	**0.010**	0.003
Inferior Occipital Gyrus	0.005	0.010	0.001	0.005	-0.002	0.007	0.000	**0.009**	-0.002	0.000	0.000	0.009
Putamen Nucleus 2	0.005	0.002	0.002	0.004	0.001	0.000	0.003	**0.005**	0.000	0.001	**0.006**	0.003

Table 1: Effect on brain correlation of removing specific categories (POS) vs. random words from training input.

Bold = p < 0.05. Given the exploratory nature of this work we did not apply the Bonferroni correction for multiple comparisons.

Some ROIs (Middle Temporal Gyrus 4, Superior Temporal Gyrus, Putamen Nucleus 3, Thalamus Nucleus) did not show any significant effect and are omitted from this table.

6 Discussion and outlook

- The fact that all the significant differences in Table 1 are positive means that eliminating specific linguistic categories damages LM-brain alignment more than randomly eliminating the same number of words, even for categories as broad as "content word" (i.e. N, Main V, Adj, Adv) or "function word", whose elimination in fact causes bigger disruption across all models.
- The elimination of Nouns and Verbs damages partially orthogonal regions (10 N-only ROIs, 2 V-only, 5 both N&V for GPT-2 L12). The same applies to Content and Function for GPT Layer-12 (1 significant ROI in common out of 13), GPT Layer-6 (1 out of 13) and LSTM (7 out of 29). When the LM is mapped onto the whole brain (Fig 1), some qualitative difference in broad regions for Nouns and Verbs appear.
- LSTM and GPT-2 layer 6 info have lower sensitivity for Ns and (to a lesser extent) Vs. In GPT-2, Noun info seems to develop after layer 6.

Future directions: (i) experimenting with different codings (here we used characters, to avoid OOV effects); (ii) use category-deprived corpora to fine-tune larger, pre-trained LMs with better correlation.

We could not come to the HSP conference due to recent changes in the US visa policies.
If you have questions, scan the QR and talk to us on zoom during poster time.

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